A Surrogate Model for Reducing the Computational Cost of Neuron Simulations

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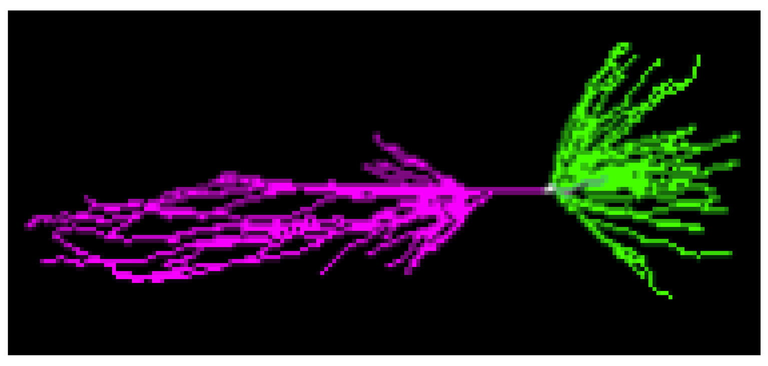
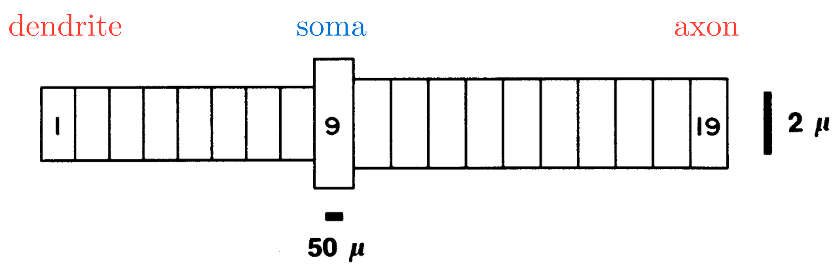
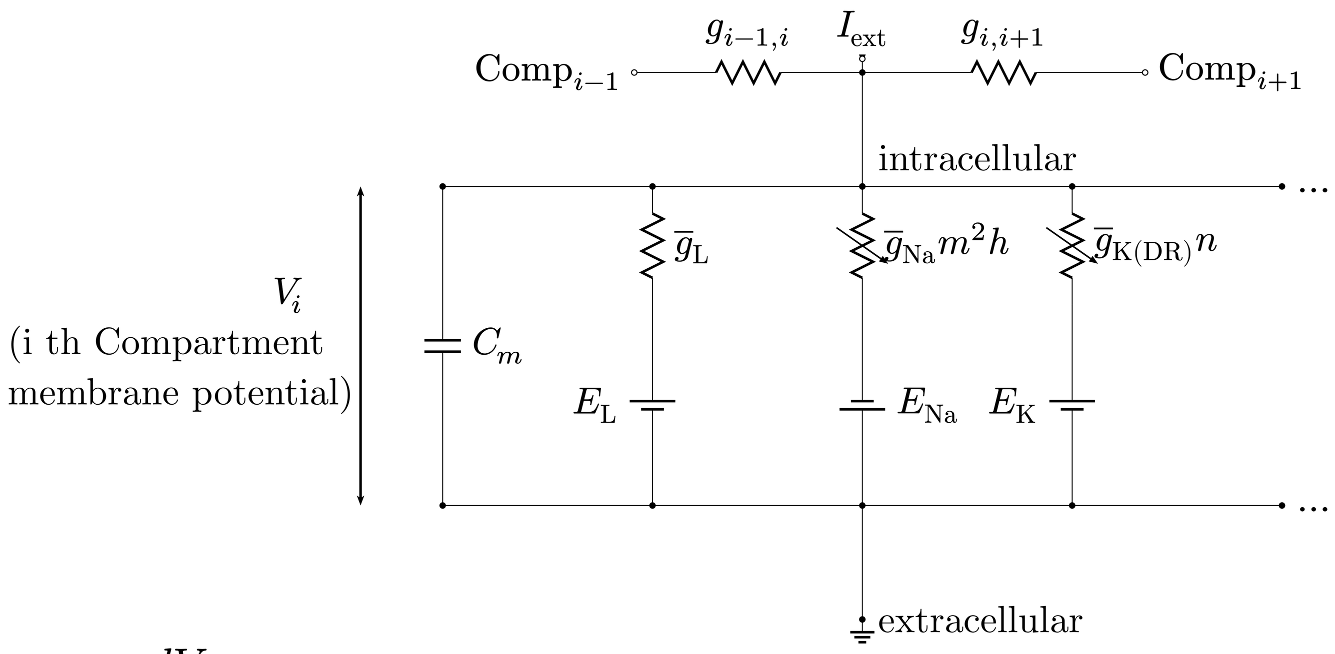
Introduction



Guinea pig CA3 pyramidal neuron[1]



Modeled as
19 compartments (comps) [2]



$$C_m \frac{dV}{dt} = -g_{\text{leak}}(V - V_{\text{leak}}) - \bar{g}_{\text{Na}} m^2 h (V - V_{\text{Na}}) - \bar{g}_{\text{K}} n (V - V_{\text{K}})$$
$$+ I_{\text{other ionic currents}} + I_{\text{ext}}$$

The conductances depend on **gate variables**, which follow ODEs with rate functions α, β:

$$I_{\text{Na}} = \bar{g}_{\text{Na}}\, m^2 h\, (V - E_{\text{Na}})$$
$$\frac{dm}{dt} = \alpha_m(V)(1-m) - \beta_m(V) m$$
$$\alpha_m(V) = 0.32(13.1 - V)/(\exp((13.1 - V)/4) - 1)$$
$$\beta_m(V) = 0.28(V - 40.1)/(\exp((V - 40.1)/5) - 1)$$

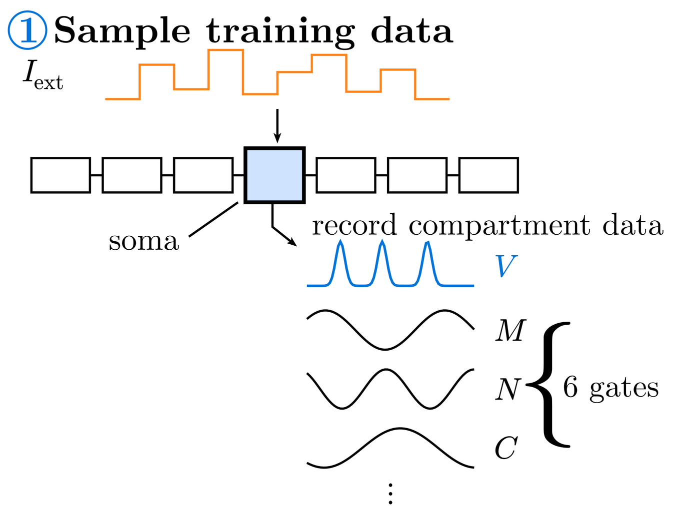
→ **11 states (10 gates and V) per comp**
→ **209** states for 19 comps;
In large scale network simulations, the number of gate variables becomes a **memory bottleneck**.

Purpose

Development of a multi-compartment neuron surrogate model capable of reproducing the membrane potential response with fewer gate variables.

Methods

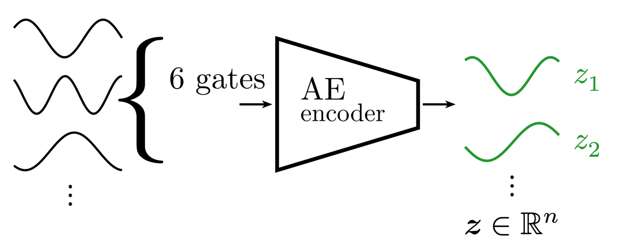
①Sample training data



Inject a **random pulse train** at the soma; record V and **6 gates**.

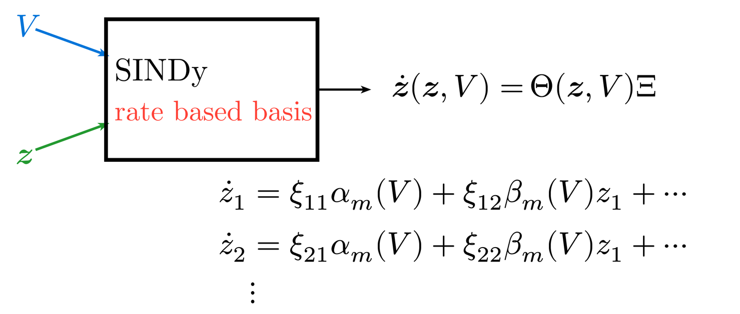
(The remaining gates, driven by Ca^{2+} dynamics, are left untouched.)

②Compress the gates



The **6 gates** are compressed to 5 latent variables by an AutoEncoder (encoder and decoder each with one hidden layer).

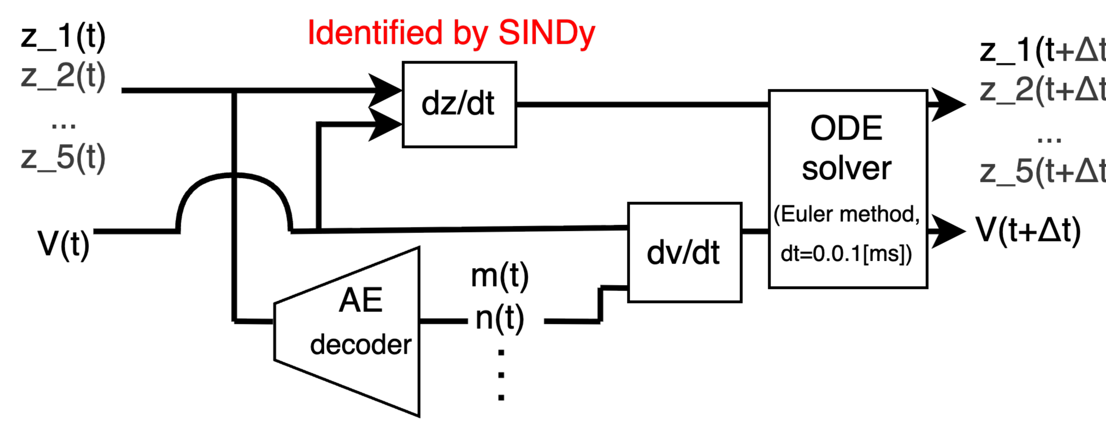
③Identify ODEs of the latent variables



Capture the latent variable dynamics with **SINDy**[3]. SINDy fits coefficients Ξ.

The library is **physics-informed**: it is built from the gates' $\alpha(V), \beta(V)$.

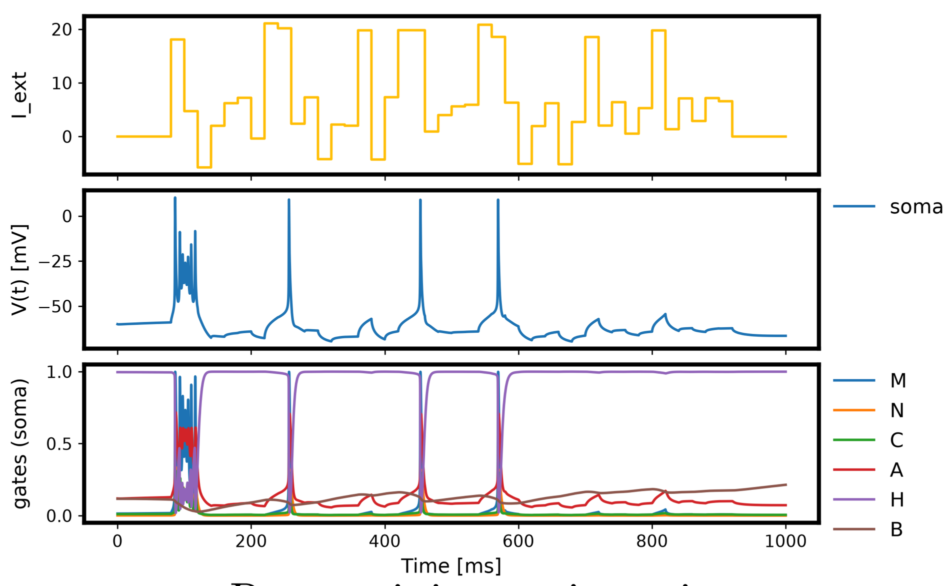
④How to apply the surrogate model



The derivative of V is computed with the **decoded gates**. Across time steps, **only the latent variables and the membrane potential need to be stored**.

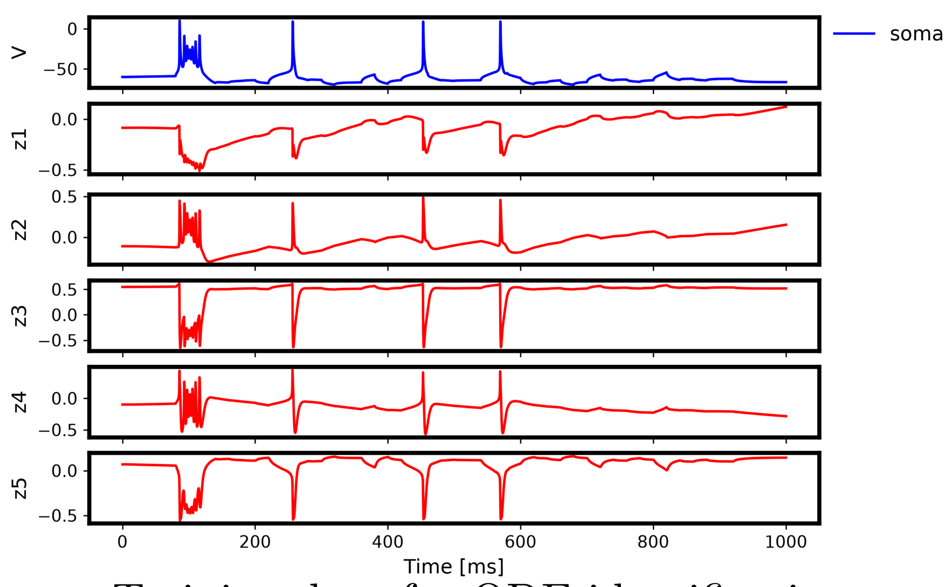
Results and Discussion

Training Data to capture gate dynamics



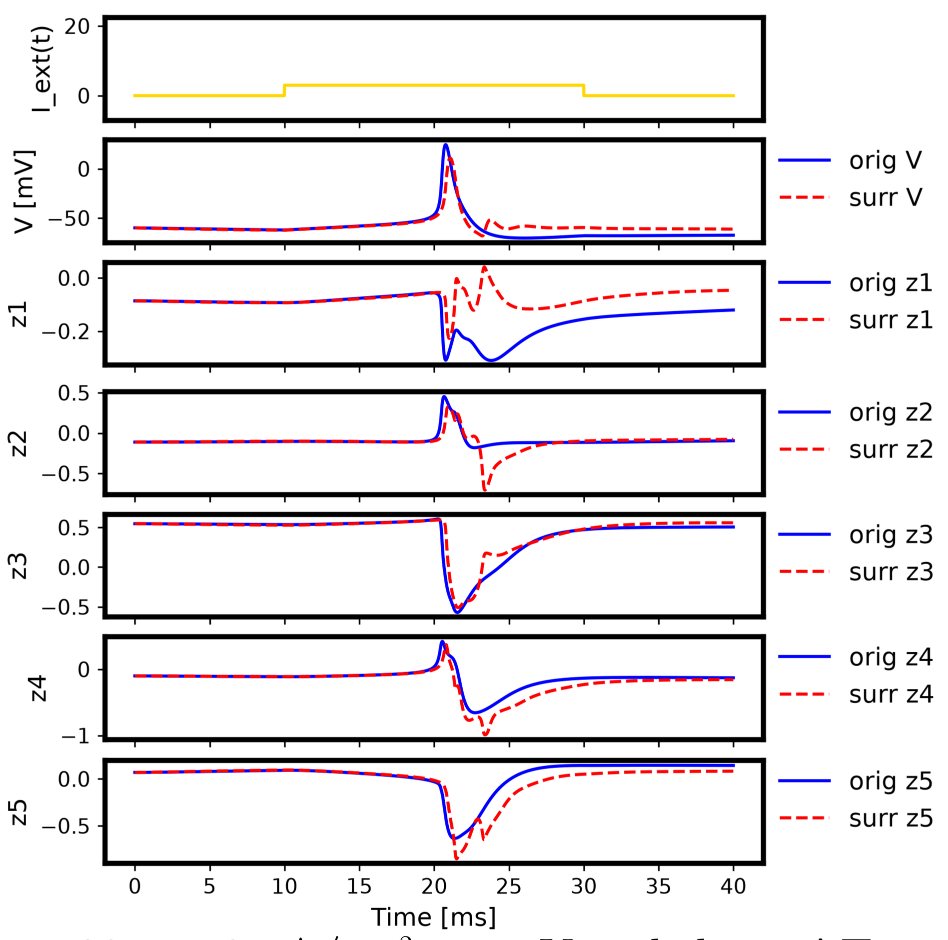
Raw training trajectories.

↓ compress gates by AutoEncoder



Training data for ODE identification.

Action Potential reproduction

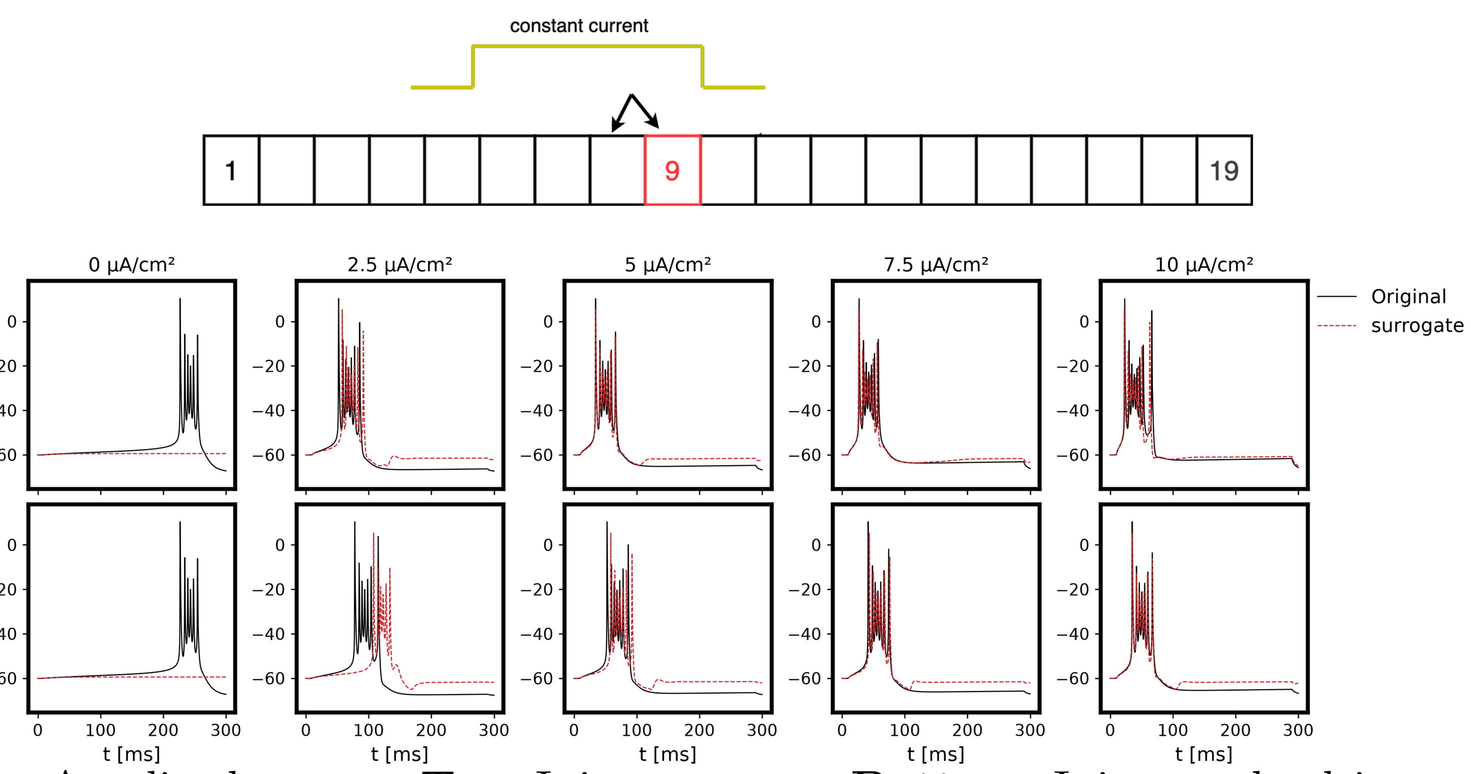


20 ms, 3 μA/cm^2 step: V and the **5 AE latents.**

- spike latency error **0.3 ms**, AHP timing gap **3.1 ms**.
- peak **13 mV low** (amplitude gap **12.9 mV**), rise / fall rate gap **21.0** / **10.3 mV/ms** — yet AHP depth gap is only **0.28** mV.

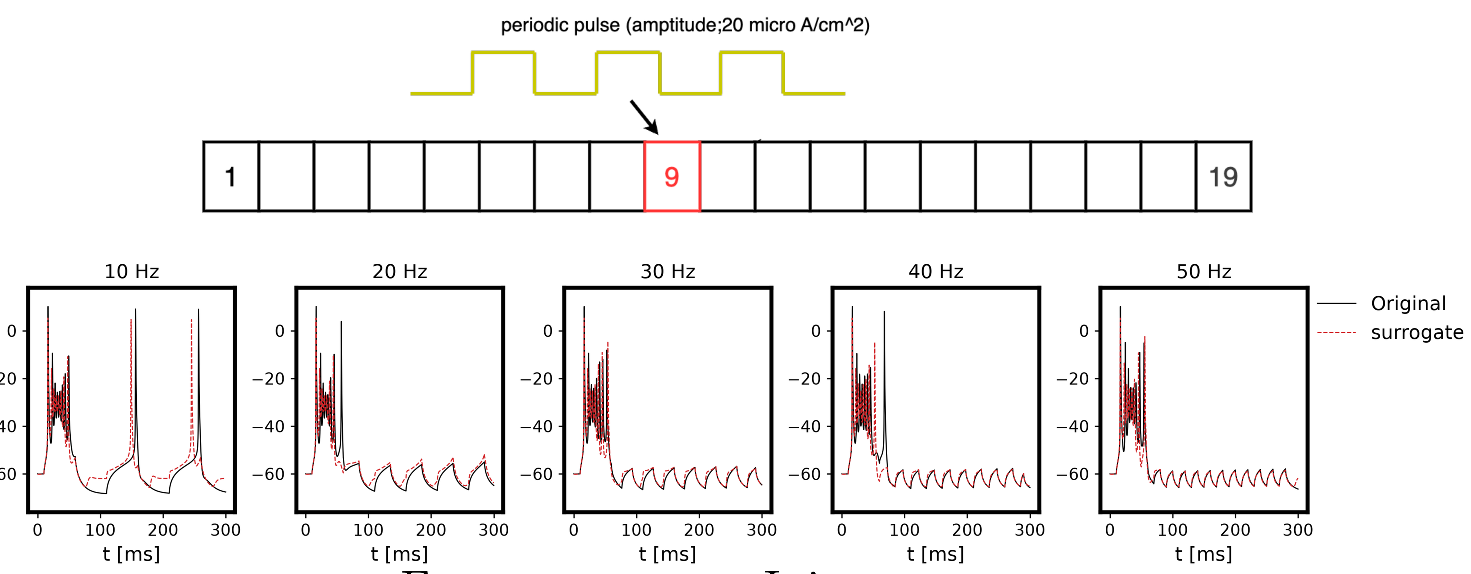
Replace the soma compartment with the surrogate model

soma compartment (9th comp) is replaced with the surrogate model.



Amplitude sweep: **Top**: Inject to soma. **Bottom**: Inject to dendrite.

- Spontaneous firing is **not reproduced. Bursts appear for $I \geq 2.5$**: delayed near threshold, but well timed for $I \geq 5$. The post-burst resting potential is **more depolarized** than the original.

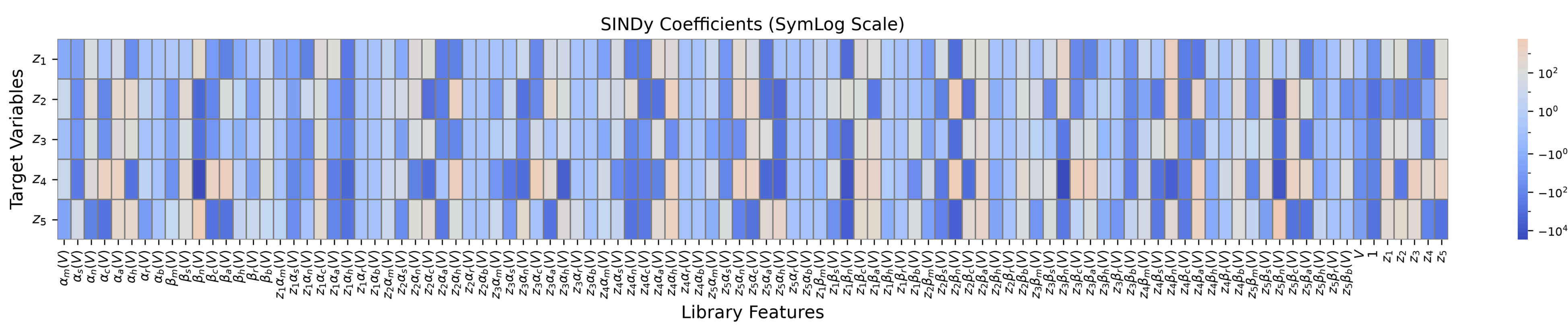


Frequency sweep: Inject to soma.

- For $f \geq 30$ Hz, the model reproduces the response well. At 10 Hz, it fires about 10 ms early.

Identified latent equations

SINDy coefficients: 79.6% non-zero



Conclusion

- The surrogate replaces the **soma** compartment only (11 → 10 states), as a first step toward a multi-compartment neuron surrogate model.
- Waveforms are **well reproduced** under the training stimulus conditions.
- Dynamics **outside the training data**, such as spontaneous firing, are **not reproduced**.

Future work

- Explore dimensionality reduction and ODE identification methods that capture the **structure of the original equations** more strongly.
- Replace **all compartments** accurately, with further reduction of gate variables.

Code — github.com/MunechikaHaruki/SINDyNeuroSurrogate

[1] [Online]. Available: https://neuromorpho.org/neuron_info.jsp?neuron_name=L16a

[2] R. D. Traub, R. K. Wong, R. Miles, and H. Michelson, "A Model of a CA3 Hippocampal Pyramidal Neuron Incorporating Voltage-Clamp Data on Intrinsic Conductances," *Journal of Neurophysiology*, vol. 66, no. 2, pp. 635–650, Aug. 1991, doi: 10.1152/jn.1991.66.2.635.

[3] K. Champion, B. Lusch, J. N. Kutz, and S. L. Brunton, "Data-Driven Discovery of Coordinates and Governing Equations," *Proceedings of the National Academy of Sciences*, vol. 116, no. 45, pp. 22445–22451, Nov. 2019, doi: 10.1073/pnas.1906995116.

Conflicts of Interest — The authors declare no conflicts of interest regarding this manuscript.